

YOLO-based Food Damage Detection: An Automated Approach for Quality Control in Food Industry

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Abstract— In modern food production and quality control, accuracy and efficiency are paramount. This study presents a pioneering application of artificial intelligence technology that integrates the advanced object recognition model YOLOv8 into food quality assessment. This approach uses high-resolution images for detection and aims to transform the food industry by providing real-time insights and actionable data to ensure quality. The main goal of this research is to create a comprehensive AI-based system that uses YOLOv8 and #039 capabilities to identify and locate spoiled food in complex and diverse production flows. Leveraging YOLOv8's state-of-the-art object detection capabilities, the approach enables rapid detection of damaged products and quality assessment, improving overall food quality control practices. In addition, YOLOv8 integration with cloud-based systems facilitates dynamic visualization and monitoring of key performance metrics. With these visualization tools, the system can provide real-time information about detection accuracy, model behavior, and potential biases. This continuous monitoring enables quick adjustments and refinements, resulting in an optimized and reliable food quality control solution. With the innovative fusion of YOLOv8 and cloud technology, the paper aims to promote accurate food quality control. By providing kitchen processors with timely and accurate information about product quality and potential damage, the system can transform traditional quality assurance practices and usher in a new era of information-driven food production.

Keywords—YOLOv8, food quality assessment, information-driven food production

I. INTRODUCTION

Traditional methods of detecting food spoilage often rely on manual inspection, which can be time-consuming, subjective and prone to human error. Therefore, there is a growing need for automatic and objective techniques for accurate detection of spoiled foods. This is where the food detection department comes in, based on the development of artificial intelligence and computer vision. Recent advances in artificial intelligence and computer vision have revolutionized the detection of food spoilage. Techniques such as deep learning and especially Convolutional Neural

Networks (CNN) have shown exceptional abilities in recognizing patterns and anomalies in images. The popular YOLO (You Only Look Once) object detection framework has been adapted for food detection. YOLO-based models can effectively analyze images or video streams and identify spoiled food with high accuracy. Quality Control in Food Processing: Food processing plants can use automated systems to inspect products for defects. This ensures that only high-quality products pass through the production line. Grocery stores and distribution centers can use food detection systems to identify damaged products on shelves, reducing waste and improving customer satisfaction.

II. LITERATURE REVIEW

In 2019, Lei Zhou, Chu Zhang, Fei Liu, Zhengjun Qiu and Yong He, in their study titled "Application of Deep Learning in Food: A Review" [1] provide a comprehensive overview of deep learning techniques applied in food quality control, emphasizing the utility of convolutional neural networks (CNNs) for object detection tasks. While the review offers valuable insights into the advantages of deep learning, such as robust feature extraction and classification, it may overlook traditional image processing methods or hybrid approaches that could also be relevant in food quality assessment. Moreover, the paper might not analyze the challenges related to data collection and annotation, crucial factors for training effective object detection models in the food industry. The authors [2] explore the domain of food image analysis using deep learning techniques in their paper titled "Multi-Scale Multi-View Deep Feature Aggregation for Food Recognition". While discussing the challenges specific to food image analysis, including lighting variations and background complexities, the study may lack in-depth coverage of object detection algorithms for enabling food quality control. Additionally, it might not provide extensive insights into recent advancements or comparative evaluations of different techniques in the context of food quality assessment.

Anuja Bhargav & Atul Bansal [6], in their review "Fruits and vegetables quality evaluation using computer vision: A review" offer a comparative analysis of various deep learning models for food quality inspection tasks, including object detection. Despite providing insights into model performance metrics such as accuracy and computational efficiency, the review might not thoroughly address real-world implementation challenges or considerations specific to food processing environments. Furthermore, it may not discuss limitations related to data scarcity or model interpretability, which are crucial in practical applications. In 2021, Xiaolei Zhang, Jie Yang, Tao Lin, Yibin Ying present "Food and agro-product quality evaluation based on spectroscopy and deep learning: A review", focusing on the latest advancements in applying deep learning to food quality control, including object detection methods [8]. While discussing topics like data augmentation and transfer learning, the study may lack in-depth examination of practical challenges or limitations encountered in real-world food processing settings. Additionally, it might not offer detailed guidance on implementing deep learning technologies effectively within the constraints of the food industry.

In a study of 2024 titled "Feature Extraction Algorithm of Food Spoilage Early Warning Data Based on Machine Learning and Internet of Things" [11], Haixia Ding explored the application of feature-based methods for detecting food spoilage. It was discussed various features extracted from food images, such as color, texture, and shape, and their significance in identifying spoiled food items. However, the study might lack in-depth analysis of specific feature extraction techniques or their effectiveness across different food types, potentially limiting its applicability in practical food quality assessment scenarios. In a study of 2017 titled [3] "Template matching for food waste detection: A review," Chen, H., & Wang, L. provided an overview of template matching technology for detecting food waste. They discussed the principles of template matching, where predefined patterns are compared with images or data of food to identify moments of deterioration or damage.

However, the review might lack in-depth analysis of the performance of template matching algorithms across different types of food products or under varying environmental conditions, potentially limiting its applicability in real-world scenarios. In a study of 2021 titled [16] "Principles and applications of convolutional neural network for spectral analysis in food quality evaluation: A review", authors explored the use of Faster CNN (Convolutional Neural Networks) architectures for accelerating food quality evaluation processes. They discussed techniques such as model pruning, quantization, and parallel computing to reduce computational complexity and speed up inference times. However, the study might lack in-depth analysis of the trade-offs between speed and accuracy in accelerated CNN models, potentially limiting its applicability in real-world food quality assessment scenarios.

Luis Castro and Sandro Augusto Magalhaes (2021)

investigated the application of deep learning techniques, including the Single Shot MultiBox Detector (SSD) [17], for automatic food waste detection. The study focused on developing a system capable of accurately identifying and segmenting food waste regions from images. By leveraging SSD's ability to predict object segments and bounding boxes directly from multiscale feature maps in a single neural network pass, the researchers aimed to achieve efficient and accurate detection of food waste. However, while SSD offered promising results in automating the detection process, the study also highlighted certain drawbacks. One of the limitations observed was related to the complexity of training SSD models, requiring large datasets and significant computational resources. Additionally, the performance of SSD could be affected by variations in lighting conditions, image quality, and the diversity of food waste types, posing challenges in real-world applications. Despite these drawbacks, the potential of SSD and other deep learning techniques in advancing food waste detection and management efforts, emphasizing the need for further research to address existing limitations and enhance system robustness. Majchrowska et al. (2022) proposed a two-stage deep learning-based framework for waste detection that consists of a detector using EfficientDet-D2 and a classifier using EfficientNet-B2 [9]. Ramanathan (2023) conducted a comprehensive review of IoT and big data-based food waste management models, algorithms, and technologies to reduce food waste from farm to fork [15].

III. PROPOSED METHODOLOGY

The proposed work aims to integrate the advanced object recognition model YOLOv8 into food quality assessment within modern food production and quality control processes. Leveraging high-resolution images for detection, the study seeks to revolutionize the food industry by offering real-time insights and actionable data to ensure quality standards are met.

The primary objective of the research is to develop a comprehensive AI-based system that harnesses YOLOv8's capabilities to identify and locate spoiled food items within complex and diverse production flows. By leveraging YOLOv8's state-of-the-art object detection capabilities, the approach enables rapid detection of damaged products and facilitates quality assessment, thereby enhancing overall food quality control practices.

In the process of developing a food leak detection system using YOLOv8, the data collection phase involved gathering around 5,000 images from various sources. These images were divided into distinct sets for testing, training, and validation to ensure that the model had a diverse and well-rounded dataset for learning. Roboflow Annotate was utilized for the annotation process, where objects within the images, such as food leaks, were manually labeled. This step required drawing bounding boxes or shapes around the areas of interest—specifically, the portions of the image where food leaks were present. Annotation is a crucial step as it provides the model with clear examples of what to identify during

training. Once the annotation process was complete, the annotated data was exported in a format compatible with YOLOv8, such as the YOLO format, which includes necessary details like class labels, bounding box coordinates, and object confidence scores.

The cloud environment setup followed, where a cloud-based development platform was established to run YOLOv8. The training process involved feeding the labeled and formatted data into YOLOv8, allowing the model to learn from the provided examples. During the training phase, progress was tracked using WandB (Weights and Biases), which provided visual insights into key metrics like loss, accuracy, and other performance indicators. This allowed developers to monitor how well the model was learning and make necessary adjustments. After training, the model was evaluated by predicting object trends in new data. The model's predictions were then compared to actual real-world data to assess its accuracy and reliability. Based on the evaluation results, the best-performing model checkpoint was selected for further use.

The trained models, along with other necessary files, were saved to the cloud, making them easily accessible for future applications. A graphical user interface (GUI) was also developed using Tkinter, providing a user-friendly platform for interaction. Within the GUI, users could upload images or videos, and the trained model would make predictions based on the media provided. The model's predictions were then displayed to the user, showcasing its ability to detect food damage with accuracy. The workflow of the proposed method is shown in figure 1.

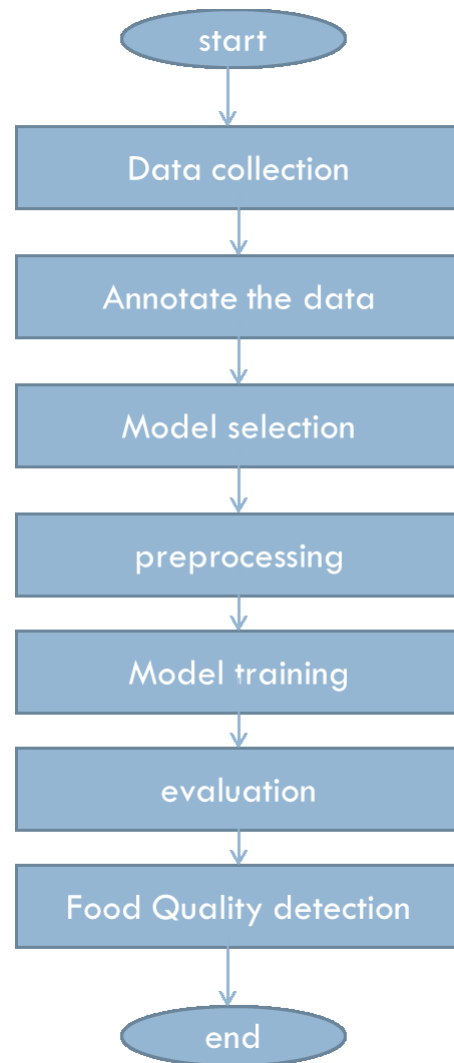


Fig. 1 Proposed Workflow

In training YOLOv8, configuration files (.yaml files) play a critical role in defining the model's architecture, training parameters, and the data paths necessary for the process. The YAML configuration allows developers to customize key elements such as input size, batch size, and the number of epochs, enabling a tailored approach to model development.

Below give is the breakdown of some of the key arguments:

img-size: This parameter specifies the input image size. It defines the dimensions to which all images are resized before being fed into the network.

batch-size: The batch size refers to the number of images processed at once during training. Adjusting this value can help manage memory usage and optimize training speed.

epochs: This determines the number of times the model will go through the entire dataset during training. A higher number of epochs typically leads to better performance but takes more time.

data: This argument provides the path to the custom .yaml configuration file, which contains details on the dataset being used, including class labels, paths to training/validation data, and any augmentation strategies.

cfg: This parameter specifies the path to the YOLOv8 model configuration file, outlining the model architecture (such as layers, filter sizes, and connections between layers).

weights: YOLOv8 allows for the use of pre-trained weights, providing a solid starting point, especially if there are existing models that were trained on similar datasets. Specifying the path to these weights helps jumpstart training and can lead to faster convergence.

During training, the system processes images based on these parameters, optimizing the weights of the model with each pass through the data.

The use case diagram of the proposed model is shown in figure 2.

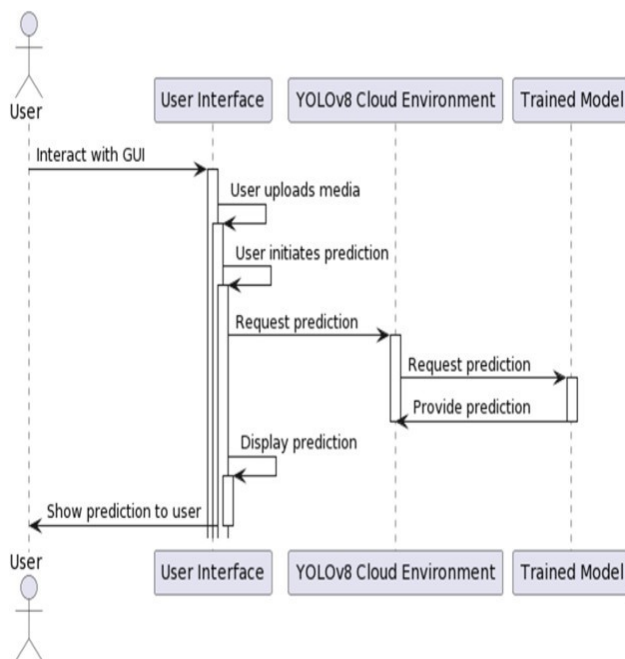


Fig. 2 Use-case Diagram

IV. RESULTS AND DISCUSSION

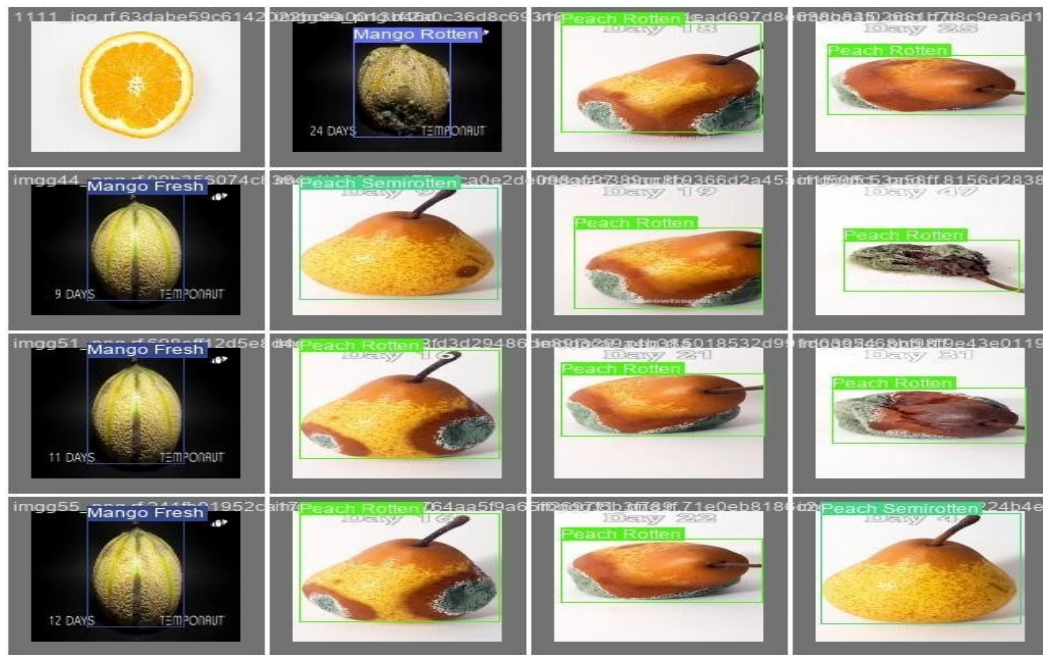


Figure 3 Detected Results

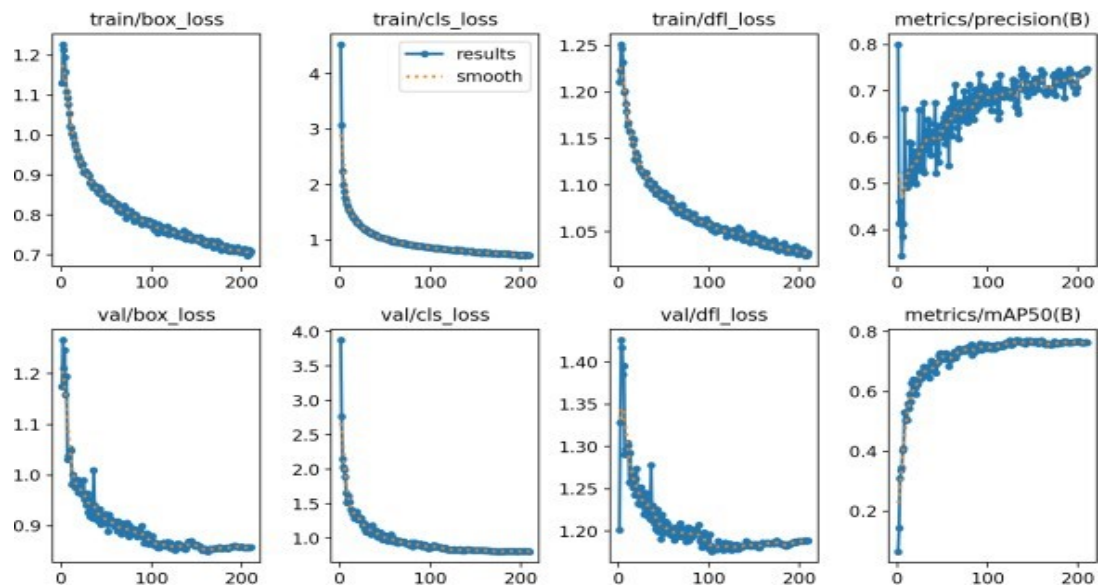


Figure 4 Performance Metrics

In figure 3, the grid of images shows various fruits (primarily mangoes and peaches) in different stages of decay, identified and labeled by a machine learning model. The labels include categories such as:

"Mango Fresh"; "Mango Rotten"; "Peach SemiRotten";
 "Peach Rotten"

Each image has a bounding box and label, which indicates the model's ability to detect and classify the fruit condition (fresh, semi-rotten, or rotten). Some images also mention the number of days the fruit has been stored, such as "24 days" for a rotten mango, demonstrating the model's focus on food spoilage over time.

Figure 4 shows a set of graphs provides insights into the performance of the machine learning model (likely a YOLO model) over time. The graphs include:

train/box_loss: The loss associated with bounding box predictions during training. The graph shows a steep decline, indicating improved accuracy.

train/cls_loss: Classification loss during training, also reducing over time, reflecting the model's ability to better classify objects.

train/dfl_loss: Loss related to the model's detection function, also showing improvement.

metrics/precision(B): The model's precision, improving steadily, suggesting fewer false positives as training progresses.

val/box_loss, val/cls_loss, val/dfl_loss: Validation losses, which are generally higher than training losses but show a similar declining trend.

metrics/mAP50(B): Mean Average Precision at 50% Intersection over Union (IoU), indicating an increase in the model's accuracy for detecting food spoilage.

These results suggest that the food spoilage detection system is improving in performance with each epoch, with a reduction in training and validation losses and an increase in precision and accuracy.

The successful integration of YOLOv8 into food quality assessment signifies a significant advancement in the field of food production and quality control. Overall, the innovative fusion of YOLOv8 and cloud technology promotes accurate food quality control practices, ultimately enhancing product quality, reducing waste, and ensuring consumer satisfaction. As technology continues to evolve, further research and development in this area will be crucial for advancing food production processes and meeting the demands of a rapidly changing market.

V. CONCLUSION

The integration of YOLOv8 into food quality assessment represents a significant advancement in modern food production practices. By leveraging the model's object detection capabilities, the system can effectively identify and locate spoiled food items, ensuring high-quality standards and minimizing product losses. The integration of YOLOv8 with cloud-based systems enhances the system's capabilities by enabling dynamic visualization, monitoring, and real-time adjustments. This fosters a culture of innovation and continuous improvement within the food industry. As technology continues to evolve, further research and development in this area will be crucial for meeting the evolving demands of the food industry and ensuring the delivery of high-quality products to consumers worldwide. The fusion of YOLOv8 and cloud technology paves the way for a new era of information-driven food production.

REFERENCES

- [1] Lei Zhou, Chu Zhang, Fei Liu, Zhengjun Qiu, Yong He, et al. Application of Deep Learning in Food: A Review. *Wiley Online Library*. (2019). doi: 10.1111/1541-4337.12492.
- [2] S. Jiang, W. Min, L. Liu and Z. Luo, "Multi-Scale Multi-View Deep Feature Aggregation for Food Recognition," in *IEEE Transactions on Image Processing*, vol. 29, pp. 265-276, 2020, doi: 10.1109/TIP.2019.2929447.
- [3] Chen, H., & Wang, L. (2017). Template matching for food waste detection: A review. *Sensors*, 21(24), 8176.
- [4] Kadus, T., Nirmal, P., & Kulkarni, K. (2020). Smart Waste Management System using IOT. *International Journal of Engineering Research and Technology*, 9(4), 1–6, doi: 10.17509/ijert.v9i4.24656.
- [5] Perez-Daniel, K., Fierro-Radilla, A., Peñaloza-Cobos, J.P. (2020). Rotten Fruit Detection Using a One Stage Object Detector. In: Martínez-Villaseñor, L., Herrera-Alcántara, O., Ponce, H., Castro-Espinoza, F.A. (eds) *Advances in Computational Intelligence*. MICAI 2020. *Lecture Notes in Computer Science*(), vol 12469. Springer, Cham. doi:10.1007/978-3-030-60887-3_29.
- [6] Bhargava, A., Bansal, A.: Fruits and vegetables quality evaluation using computer vision: a review. *Journal of King Saud University - Computer and Information Sciences*, pp. 1–15 (2021)
- [7] Gull, S., Bajwa, I. S., Anwar, W., & Rashid, R. (2021). IoT-based smart garbage system for efficient food waste management. *The Research Article*, 2021, 1–6, doi: 10.17509/ij-srist.2021.9.1.1.
- [8] Xiaolei Zhang, Jie Yang, Tao Lin, Yibin Ying, Food and agro-product quality evaluation based on spectroscopy and deep learning: A review, *Trends in Food Science & Technology*, Volume 112, 2021, Pages 431-441, ISSN 0924-2244, doi: 10.1016/j.tifs.2021.04.008.
- [9] Majchrowska, S., Mikołajczyk, A., Ferlin, M., Klawikowska, Z., Kwasigroch, A., & Majek, K. (2022). Deep learning-based waste detection in natural and urban environments. *Waste Management*, 138, 274–284.
- [10] Lubura, J., Pezo, L., Sandu, M.A., Voronova, V., Donsi, F., Šic Žlabur, J., Ribić, B., Peter, A., Šurić, J., Brandić, I., et al. (2022). Food Recognition and Food Waste Estimation Using Convolutional Neural Network. *Electronics*, 11(22), 3746, doi: 10.3390/electronics11223746.
- [11] Ding, H. (February 16, 2024). "Feature Extraction Algorithm of Food Spoilage Early Warning Data Based on Machine Learning and Internet of Things." *ASTM International. J. Test. Eval.* doi:10.1520/JTE20230066.
- [12] Naveen Kumar Mahanti, R. Pandiselvam, Anjineyulu Kothakota, Padma Ishwarya S., Subir Kumar Chakraborty, Manoj Kumar, Daniel Cozzolino, Emerging non-destructive imaging techniques for fruit damage detection: Image processing and analysis, *Trends in Food Science & Technology*, Volume 120, 2022, doi.org/10.1016/j.tifs.2021.12.021
- [13] Nasirahmadi, A.; Wilczek, U.; Hensel, O. Sugar Beet Damage Detection during Harvesting Using Different Convolutional Neural Network Models. *Agriculture* 2021, 11, 1111. doi:10.3390/agriculture1111111
- [14] A. Osipov et al., "Identification and Classification of Mechanical Damage During Continuous Harvesting of Root Crops Using Computer Vision Methods," in *IEEE Access*, vol. 10, pp. 28885-28894, 2022, doi: 10.1109/ACCESS.2022.3157619.
- [15] Ramanathan, R.; Duan, Y.; Ajmal, T.; Pelc, K.; Gillespie, J.; Ahmadzadeh, S.; Condell, J.; Hermens, I.; Ramanathan, U. Motivations and Challenges for Food Companies in Using IoT Sensors for Reducing Food Waste: Some Insights and a Road Map for the Future. *Sustainability* 2023, 15, 1665. Doi.org/10.3390/su15021665
- [16] Na Luo, Daming Xu, Bin Xing, Xinting Yang, Chuanheng Sun, Principles and applications of convolutional neural network for spectral analysis in food quality evaluation: A review, *Journal of Food Composition and Analysis*, Volume 128, 2024, 105996, ISSN 0889-1575, doi.org/10.1016/j.jfca.2024.105996.
- [17] Magalhães, S.A.; Castro, L.; Moreira, G.; dos Santos, F.N.; Cunha, M.; Dias, J.; Moreira, A.P. Evaluating the Single-Shot MultiBox Detector and YOLO Deep Learning Models for the Detection of Tomatoes in a Greenhouse. *Sensors* 2021, 21, 3569. doi.org/10.3390/s21103569